

Vector Differentiation: Scalar Potential of an Irrotational Field

1. Scalar Potential of a Conservative Field

Name of Concept

Conservative Vector Fields and Finding the Scalar Potential (ϕ).

Core Mathematical Formula

If a vector field $\vec{F}$ is irrotational ($\nabla \times \vec{F} = \vec{0}$), it can be expressed as the gradient of a scalar function ϕ . This scalar function ϕ is called the **scalar potential**:

$$\vec{F} = \nabla \phi$$

This vector equation expands into three independent scalar differential equations:

$$\frac{\partial \phi}{\partial x} = f_1, \quad \frac{\partial \phi}{\partial y} = f_2, \quad \frac{\partial \phi}{\partial z} = f_3$$

Beginner-Friendly Explanation of Operations

- What is a Scalar Potential?** Think of a vector field $\vec{F}$ as a force field (like gravity or an electric field). If the field is irrotational, it means moving an object along a closed loop costs zero net energy. Because energy is conserved, we can describe the field using a single underlying "hill" or "altitude map" called the scalar potential ϕ (similar to potential energy).
- Reversing the Derivative (Integration):** The field components f_1 , f_2 , and f_3 are the partial derivatives of our mystery function ϕ . To recover ϕ , we must perform partial integration:
 - Integrate f_1 with respect to x . When doing this, any term containing *only* y or z acts like a constant and disappears. So, instead of a standard constant $+C$, we add an unknown function $+f(y, z)$.
 - Integrate f_2 with respect to y , adding an unknown function $+g(x, z)$.
 - Integrate f_3 with respect to z , adding an unknown function $+h(x, y)$.
- Merging the Results:** Finally, we look at all three integration results and combine them logically. We include every unique term exactly once, along with a final constant number $+C$.

2. Step-by-Step Problem Solution

Question

A vector field is given by $\vec{F} = (x^2 + xy^2)\vec{i} + (y^2 + x^2y)\vec{j}$. Show that $\vec{F}$ is irrotational and find its scalar potential.

Step 1: Extract the vector components

Identify the scalar function components f_1 , f_2 , and f_3 :

$$f_1 = x^2 + xy^2$$

$$f_2 = y^2 + x^2y$$

$$f_3 = 0$$

Step 2: Set up and evaluate the curl determinant

$$\nabla \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (x^2 + xy^2) & (y^2 + x^2y) & 0 \end{vmatrix}$$

Expand the determinant component by component:

$$\text{Component } \bar{i} = \bar{i} \left[\frac{\partial}{\partial y}(0) - \frac{\partial}{\partial z}(y^2 + x^2y) \right] = \bar{i}[0 - 0] = \bar{0}$$

$$\text{Component } \bar{j} = -\bar{j} \left[\frac{\partial}{\partial x}(0) - \frac{\partial}{\partial z}(x^2 + xy^2) \right] = -\bar{j}[0 - 0] = \bar{0}$$

$$\text{Component } \bar{k} = \bar{k} \left[\frac{\partial}{\partial x}(y^2 + x^2y) - \frac{\partial}{\partial y}(x^2 + xy^2) \right] = \bar{k}[2xy - 2xy] = \bar{0}$$

Combine the components:

$$\nabla \times \bar{F} = 0\bar{i} - 0\bar{j} + 0\bar{k} = \bar{0}$$

Since $\nabla \times \bar{F} = \bar{0}$, the vector field $\bar{F}$ is proven to be **irrotational**.

Step 3: Set up the partial differential equations for ϕ

Since the field is irrotational, set $\bar{F} = \nabla\phi$:

$$\frac{\partial\phi}{\partial x} = x^2 + xy^2 \quad \text{--- (Equation 1)}$$

$$\frac{\partial\phi}{\partial y} = y^2 + x^2y \quad \text{--- (Equation 2)}$$

Step 4: Integrate Equation 1 with respect to x

Integrate both terms with respect to x , treating y as a constant factor. Add a placeholder function of y , labeled $f(y)$, as the constant of integration:

$$\begin{aligned} \phi &= \int (x^2 + xy^2) dx \\ \phi &= \frac{x^3}{3} + \frac{x^2y^2}{2} + f(y) \quad \text{--- (Expression A)} \end{aligned}$$

Step 5: Integrate Equation 2 with respect to y

Integrate both terms with respect to y , treating x as a constant factor. Add a placeholder function of x , labeled $g(x)$, as the constant of integration:

$$\begin{aligned} \phi &= \int (y^2 + x^2y) dy \\ \phi &= \frac{y^3}{3} + \frac{x^2y^2}{2} + g(x) \quad \text{--- (Expression B)} \end{aligned}$$

Step 6: Compare and merge expressions to find the final scalar potential

Compare **Expression A** and **Expression B** to reconstruct the complete function ϕ :

- The shared term $\frac{x^2y^2}{2}$ appears in both calculations. We write it down once.
- The term $\frac{x^3}{3}$ from Expression A matches the missing function $g(x)$ in Expression B.
- The term $\frac{y^3}{3}$ from Expression B matches the missing function $f(y)$ in Expression A.

Combine all unique terms and add a final absolute integration constant C :

$$\phi = \frac{x^3}{3} + \frac{x^2y^2}{2} + \frac{y^3}{3} + C$$